

VC_BIT

Overview

VC_BIT is a collection of four small verification components, used for the single-bit signal conversions. These conversions include the following:

- Rising edge extraction (vc_rose)
- Falling edge extraction (vc_fell)
- Changing extraction (vc_change)
- Parametrical bit delay (vc_delay)

These components are instrumental for building a more complex checking, coverage and similar expressions.

Syntax

```
vc_rose          InstName (clk, rst_n, bit, bit_rose);
vc_fell          InstName (clk, rst_n, bit, bit_fell);
vc_change        InstName (clk, rst_n, bit, bit_change);
vc_delay #(num_clocks) InstName (clk, rst_n, bit, bit_delay);
```

Interface	
clk	Clock
rst_n	Active-low reset
bit	One-bit data input
bit_rose	One-bit data output; generates pulse high when din rises
bit_fell	One-bit data output; generates pulse high when din falls
bit_rose	One-bit data output; generates pulse high when din changes
bit_delay	One-bit data output; presents delayed for the <num_clock> input

Example

```
module test;

reg a;
reg clk, rst_n;
initial begin
    clk = 1;
    rst_n = 0;
    a = 0;
end

always #5 clk <= ~clk;

always @(posedge clk)
    if ({$random}%10 < 2) a <= 1;
    else a <= 0;

vc_rose      rose_a      (clk, rst_n, a, a_rose);
vc_fell      fell_a      (clk, rst_n, a, a_fell);
vc_changed   changed_a   (clk, rst_n, a, a_changed);
vc_delay #3 delay_a      (clk, rst_n, a, a_delay);

initial begin
    $dumpvars;
    @(posedge clk);@(posedge clk);
    rst_n = 1;

    #500 $finish;
end

endmodule
```